

CUXHAVEN NORDHOLZ, Germany

WMO: 101360

Lat: 53.7647N

Lon: 8.6583E

Elev: 107

StdP: 14.64

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.1	20.7	12.2	10.3	19.3	16.0	12.4	21.8	33.8	44.1	30.2	44.0	8.1	100	0.618

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.5	83.5	66.9	79.1	65.5	75.3	63.8	68.7	79.5	67.0	76.6	65.2	73.3	9.5	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.8	92.2	72.3	63.1	86.9	70.8	61.7	82.5	69.2	32.8	79.4	31.4	76.6	30.1	73.0	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.7	23.2	20.6	DB	12.5	90.3	5.6	3.0	8.4	92.4	5.1	94.2	2.0	95.8	-2.1	98.0	(4)
(5)			WB	12.0	72.7	5.4	1.9	8.1	74.1	4.9	75.2	1.8	76.2	-2.2	77.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.2	35.3	36.1	40.3	47.7	54.3	59.5	64.1	64.0	58.2	50.6	42.6	37.1	(6)	
	DBStd	12.09	7.67	7.18	6.34	6.43	6.10	5.46	5.47	4.92	4.72	5.89	6.15	7.56	(7)	
	HDD50	1993	457	389	305	120	24	0	0	0	1	65	232	400	(8)	
	HDD65	5897	922	808	766	520	336	184	87	79	211	447	673	864	(9)	
	CDD50	1705	0	1	4	50	157	284	436	434	247	83	8	1	(10)	
	CDD65	136	0	0	0	0	5	18	58	48	7	0	0	0	(11)	
	CDH74	1099	0	0	0	6	59	164	477	347	46	0	0	0	(12)	
	CDH80	288	0	0	0	0	7	39	138	99	5	0	0	0	(13)	
(14) Wind	WSAvg	10.1	11.4	11.3	11.4	10.0	9.8	9.7	9.3	8.7	9.0	9.7	9.9	10.7	(14)	
(15) Precipitation	PrecAvg	33.6	2.6	2.0	2.0	1.6	2.2	3.3	3.6	3.6	3.3	3.2	3.1	3.0	(15)	
	PrecMax	44.9	5.5	4.6	3.7	3.8	4.6	7.2	7.2	8.5	7.9	7.9	6.5	7.4	(16)	
	PrecMin	19.6	0.0	0.7	0.4	0.2	0.5	0.3	1.5	1.1	1.1	1.1	0.4	0.7	(17)	
	PrecStd	5.6	1.3	1.0	1.0	0.8	1.0	1.7	1.7	1.8	1.8	1.7	1.5	1.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.2	53.8	62.2	73.4	80.2	84.7	89.4	87.7	79.2	68.4	59.1	53.9	(19)	
		MCWB	49.0	47.1	51.2	57.4	64.0	67.3	68.9	67.7	65.7	60.6	55.4	51.1	(20)	
	2%	DB	49.5	50.3	55.8	68.1	74.8	78.8	84.0	82.6	73.8	64.1	55.5	51.7	(21)	
		MCWB	47.1	46.4	48.5	54.7	60.4	64.9	67.3	66.6	63.8	58.8	52.9	49.3	(22)	
	5%	DB	46.8	47.9	51.9	63.8	70.1	73.6	79.2	77.2	69.5	61.0	53.3	48.6	(23)	
		MCWB	44.9	45.3	47.0	53.3	58.5	62.8	66.0	65.4	61.3	56.8	50.6	46.8	(24)	
	10%	DB	44.8	45.7	49.1	59.0	65.9	69.5	75.0	73.3	65.9	58.8	50.9	46.5	(25)	
		MCWB	43.2	43.3	45.5	51.2	56.5	60.7	64.4	63.6	59.5	55.4	48.5	44.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.9	49.7	53.0	59.2	66.5	69.8	71.3	71.2	67.2	61.6	56.3	52.2	(27)	
		MCDB	51.7	51.9	58.9	70.0	76.9	81.3	83.4	81.7	76.5	67.0	58.4	53.7	(28)	
	2%	WB	47.3	47.0	50.1	56.3	62.5	66.5	69.0	68.3	64.4	59.1	53.3	49.6	(29)	
		MCDB	48.8	49.3	54.1	65.2	71.3	76.4	80.3	78.0	71.9	63.2	55.4	51.1	(30)	
	5%	WB	45.0	45.2	47.8	54.1	59.4	63.8	67.2	66.4	62.1	57.4	50.9	47.1	(31)	
		MCDB	46.4	47.5	51.5	61.7	67.5	71.3	77.3	74.9	67.9	60.5	52.8	48.6	(32)	
	10%	WB	43.1	43.1	45.6	51.9	57.1	61.6	65.2	64.7	60.3	55.7	48.9	44.9	(33)	
		MCDB	44.5	45.1	48.5	58.2	64.6	67.9	73.4	72.1	64.6	58.5	50.5	46.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	8.9	11.4	15.1	15.4	14.3	14.5	14.6	12.9	11.0	8.2	7.3	(35)	
		MCDBR	8.4	10.7	16.7	23.4	23.6	22.3	23.1	22.0	18.9	13.1	8.8	8.1	(36)	
	5% WB	MCWBR	7.5	8.3	10.9	12.7	12.5	11.8	10.3	10.3	10.6	8.9	7.3	7.5	(37)	
		MCWBR	7.9	9.1	14.8	20.1	20.7	19.9	20.6	18.7	16.7	12.3	8.5	7.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.326	0.345	0.374	0.407	0.401	0.400	0.416	0.410	0.386	0.362	0.339	0.318	(40)	
		taud	2.313	2.329	2.270	2.235	2.300	2.327	2.295	2.321	2.375	2.399	2.350	2.306	(41)	
	Ebn at Noon		199	236	253	260	269	270	262	256	248	227	192	176	(42)	
		Edh at Noon	19	25	33	39	38	38	38	36	30	23	18	16	(43)	
(44) All-Sky Solar Radiation	RadAvg		176	387	771	1279	1605	1673	1592	1313	947	523	221	132	(44)	
	RadStd		20	53	76	151	171	135	206	97	110	57	23	10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-238	N/A	N/A	N/A

Nomenclature: See separate page